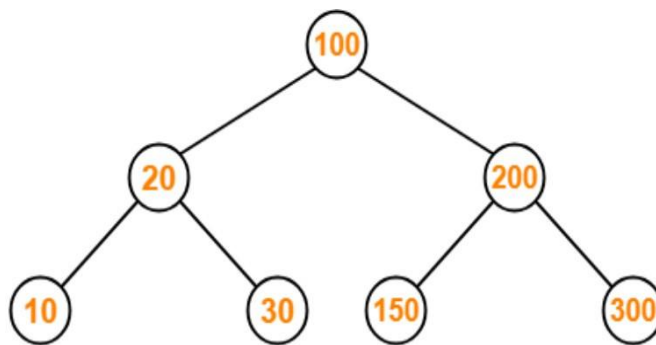


CSA0305 – Data Structures (Slot A)

C03 - AT2 - Tree Traversal Challenges

Question No:1

Write the traversal sequences for following trees

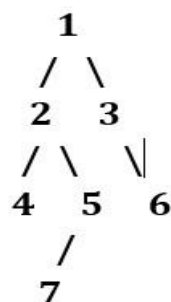


Question No: 2

Assume the digits 7, 5, 1, 8, 3, 6, 0, 9, 4, 2 are added in that sequence into a binary search tree that is initially empty. The binary search tree follows the standard natural number ordering. What is the resulting tree's traversal sequence?

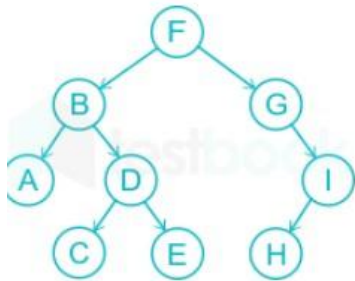
Question No: 3

Write the Inorder, Preorder & Postorder traversal sequences for following trees



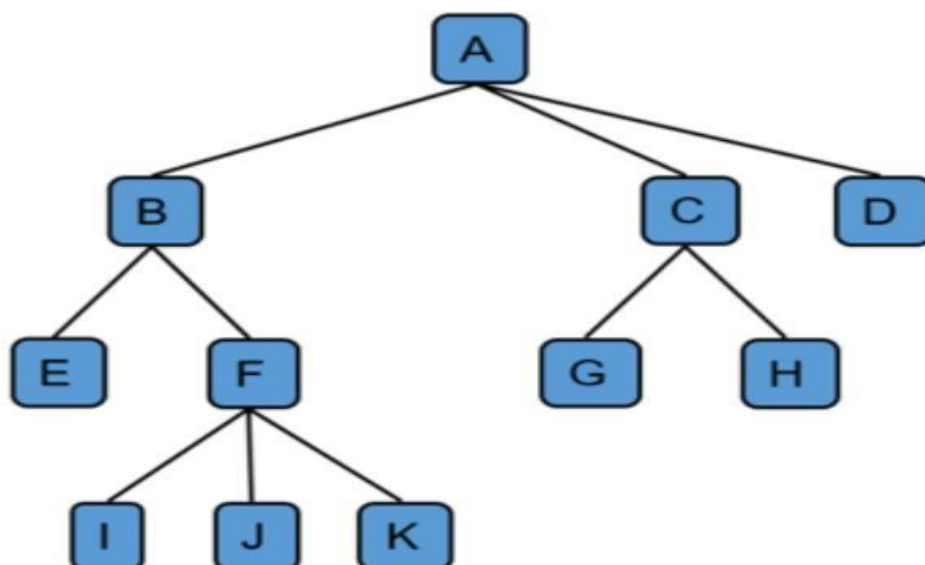
Question No: 4

What is the sequence of nodes when applying in-order traversal on the binary tree given below?



1. A, B, C, D, E, F, I, H, G
2. A, C, D, E, B, F, G, H, I
3. A, B, C, D, E, F, G, H, I
4. I, H, G, F, E, D, C, B, A

Question No: 5



QUESTION 1

PREORDER

[← Back to Home](#)

Tree Traversals

?

Input

Input is level-by-level order.

100,20,200,10,30,150,300

Enter numbers separated by commas. Use null for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

© 2025 Saifullah Bin Yusuf. All rights reserved.

[← Back to Home](#)

Tree Traversals

?

Input

Input is level-by-level order.

100,20,200,10,30,150,300

Enter numbers separated by commas. Use null for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

100

20

200

10

30

150

300

© 2025 Saifullah Bin Yusuf. All rights reserved.

[← Back to Home](#)

Tree Traversals

?

Input

Input is level-by-level order.

100,20,200,10,30,150,300

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

100201030200150300

```
graph TD; 100((100)) --- 20((20)); 100 --- 200((200)); 20 --- 10((10)); 20 --- 30((30)); 200 --- 150((150)); 200 --- 300((300)); style 300 fill:#f96
```

© 2025 Saifullah Bin Yusuf. All rights reserved.

Traversal result: [10, 20, 10, 30, 200, 150, 300]

OKAY

INORDER

[← Back to Home](#)

Tree Traversals

?

Input

Input is level-by-level order.

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

```

graph TD
    10((10)) --> 20((20))
    10 --> 200((200))
    20 --> 10_1((10))
    20 --> 30((30))
    200 --> 150((150))
    200 --> 300((300))

```

← Back to Home

Tree Traversals

?

Input

Input is level-by-level order.

100,20,200,10,30,150,300

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

10 20 30 100 150 200 300

```
graph TD; 100((100)) --- 20((20)); 100 --- 200((200)); 20 --- 10((10)); 20 --- 30((30)); 200 --- 150((150)); 200 --- 300((300)); style 300 fill:#ff9900
```

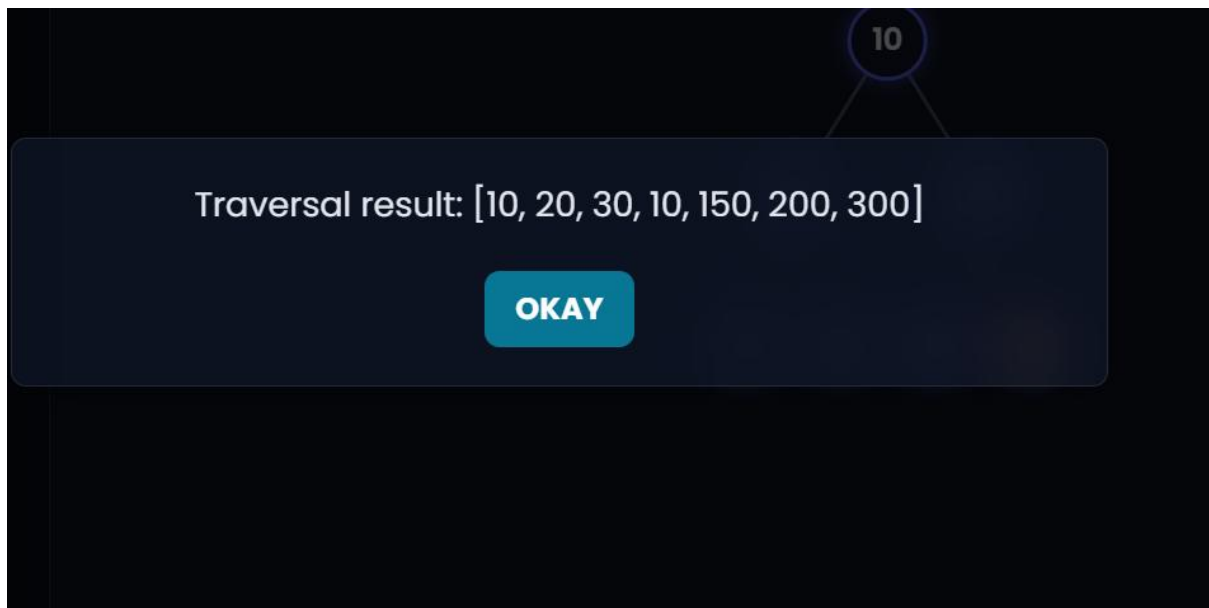
© 2025 Saifullah Bin Yusuf. All rights reserved.

🔄

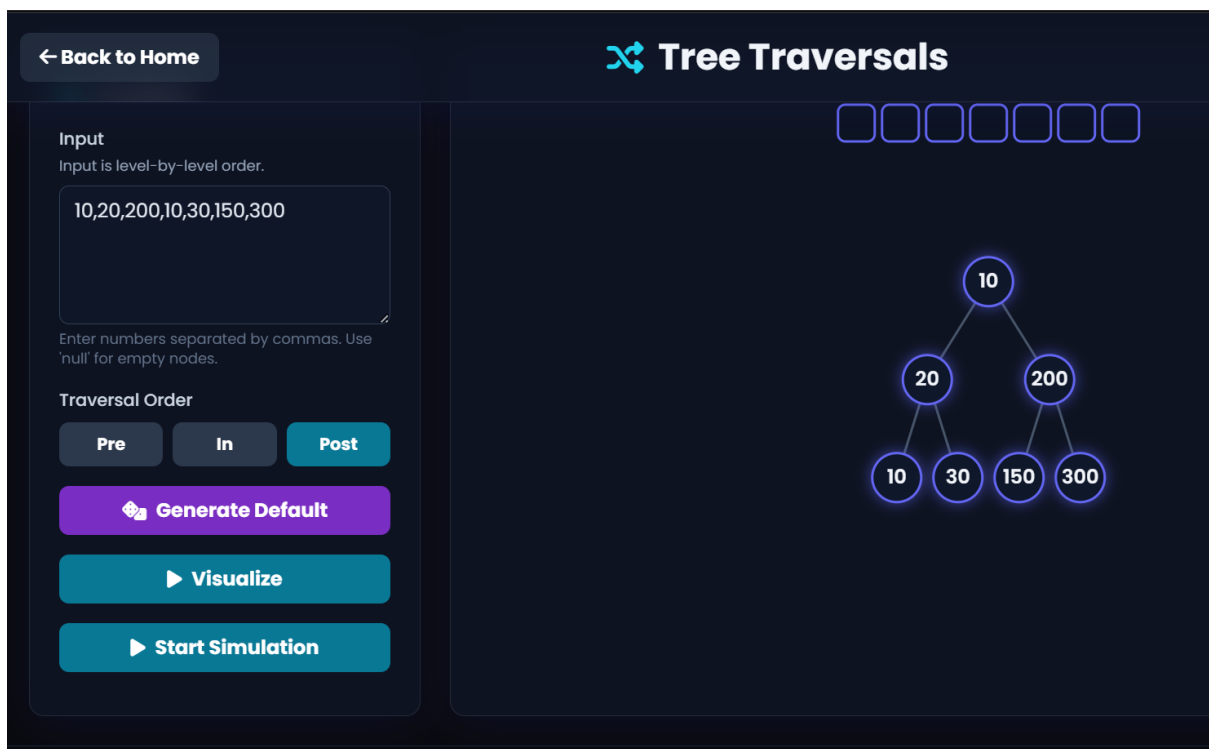
🌐

</>

✉️



POST ORDER



[← Back to Home](#)

Tree Traversals

Input

Input is level-by-level order.

100,20,200,10,30,150,300

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

10 30 20 150 300 200 100

100

20

200

10

30

150

300

© 2025 Saifullah Bin Yusuf. All rights reserved.



10

Traversal result: [10, 30, 20, 150, 300, 200, 10]

OKAY

QUESTION 2

PREORDER

Input

Input is level-by-level order.

7, 5, 8, 1, 6, null, 9, 0, 3, null, null,
null, null, 2, 4

Enter numbers separated by commas. Use
'null' for empty nodes.

Traversal Order

Pre

In

Post

 Generate Default

 Visualize

 Start Simulation

© 2025 Saifullah Bin Yusuf. All rights reserved.



← Back to Home

Tree Traversals

?

Input

Input is level-by-level order.

7, 5, 8, 1, 6, null, 9, 0, 3, null, null, null, 2, 4

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

7

5

8

1

6

9

0

3

2

4

© 2025 Saifullah Bin Yusuf. All rights reserved.

in

</>

[← Back to Home](#)

Tree Traversals

Input

Input is level-by-level order.

7, 5, 8, 1, 6, null, 9, 0, 3, null, null, null, null, 2, 4

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

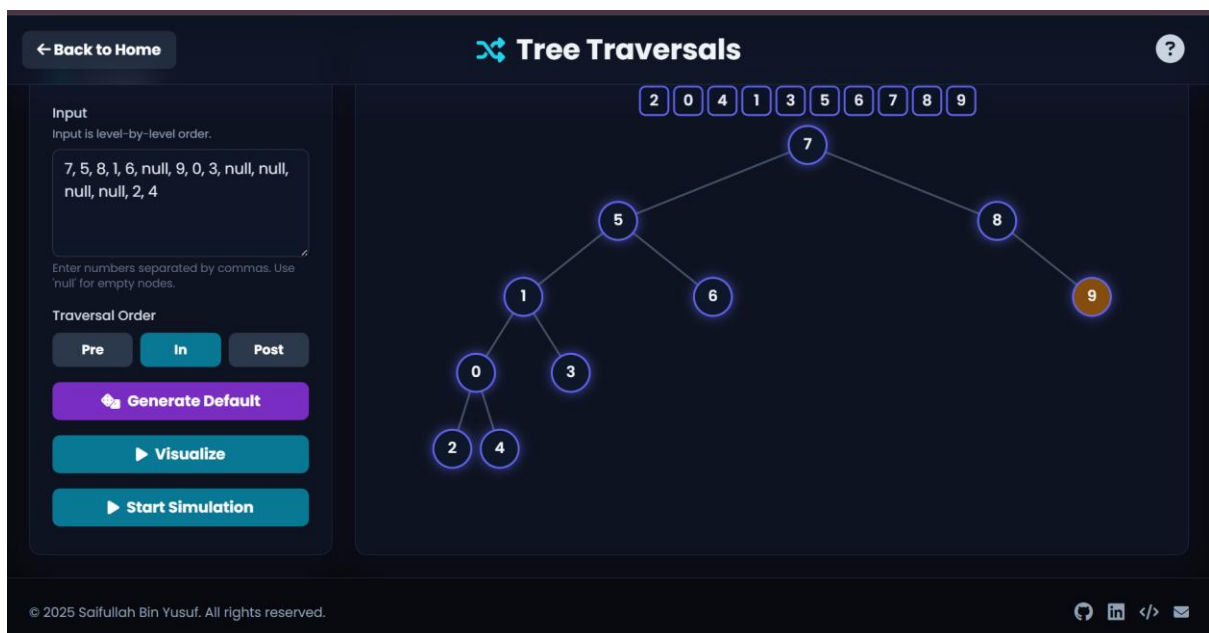
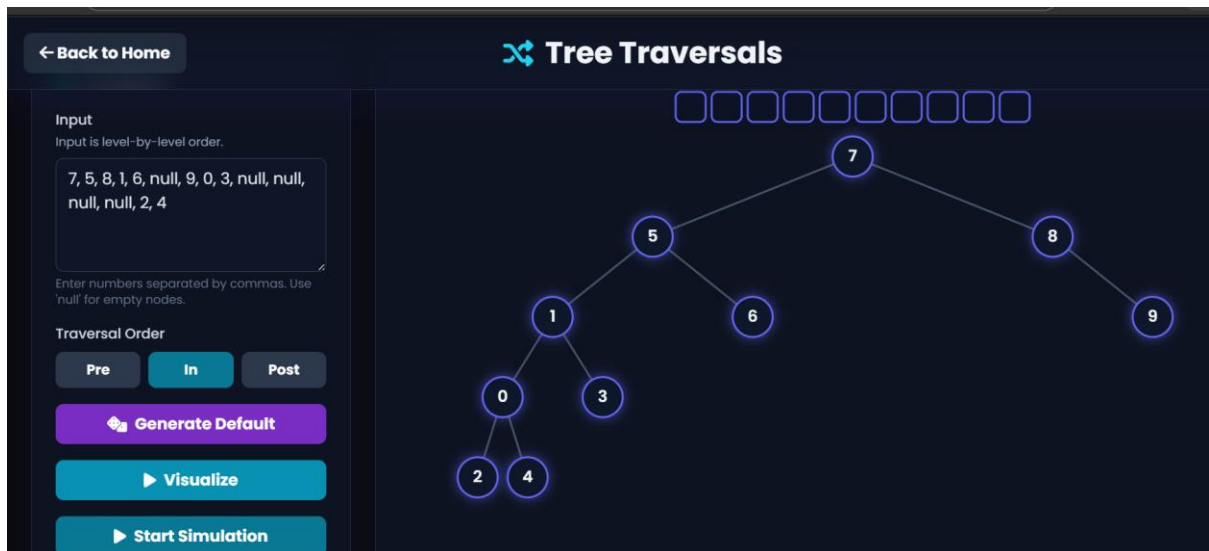
2403165987

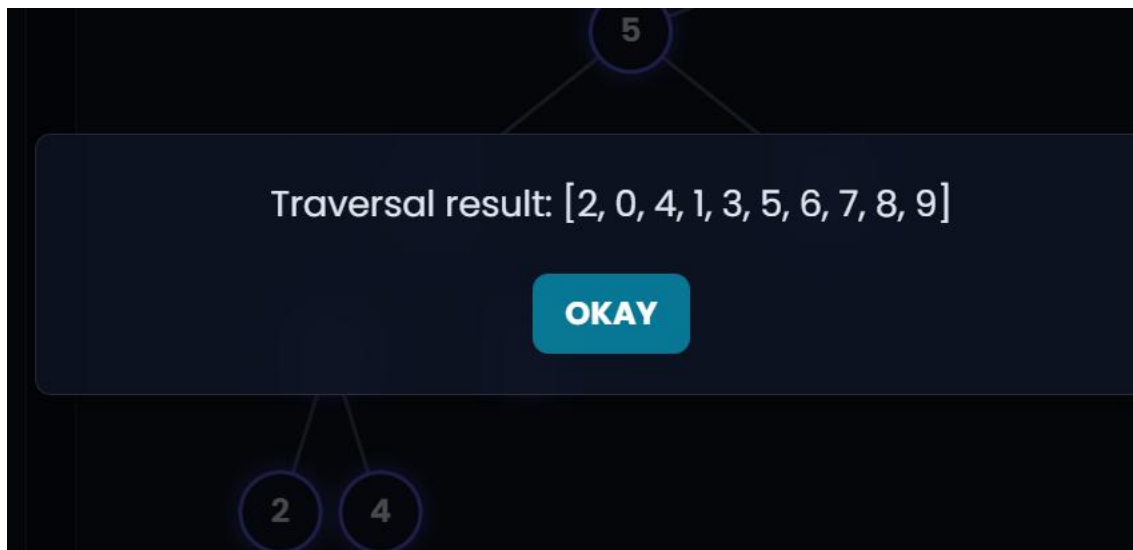
© 2025 Saifullah Bin Yusuf. All rights reserved.

Traversal result: [7, 5, 1, 0, 2, 4, 3, 6, 8, 9]

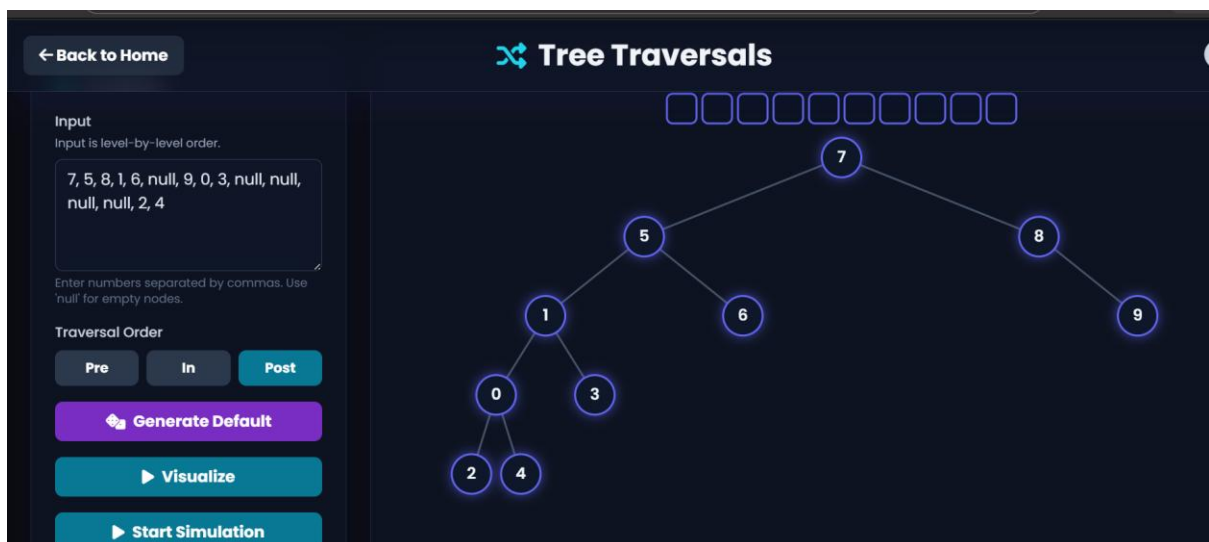
OKAY

INORDER





POSTORDER



[← Back to Home](#)

Tree Traversals

Input

Input is level-by-level order.

7, 5, 8, 1, 6, null, 9, 0, 3, null, null, null, null, 2, 4

Enter numbers separated by commas. Use null for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

2403165987

7

5

8

1

6

9

0

3

2

4

© 2025 Saifullah Bin Yusuf. All rights reserved.

5

Traversal result: [2, 4, 0, 3, 1, 6, 5, 9, 8, 7]

OKAY

2

4

QUESTION 3

PREORDER

← Back to Home

Tree Traversals

?

Input

Input is level-by-level order.

1, 2, 3, 4, 5, null, 6, null, null, 7,
null, null, null, null, null

Enter numbers separated by commas. Use
null for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

1

2

3

4

5

6

7

© 2025 Saifullah Bin Yusuf. All rights reserved.

↺

in

</>

✉

← Back to Home

Tree Traversals

?

Input

Input is level-by-level order.

1, 2, 3, 4, 5, null, 6, null, null, 7, null, null, null, null, null

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

1 2 4 5 7 3 6

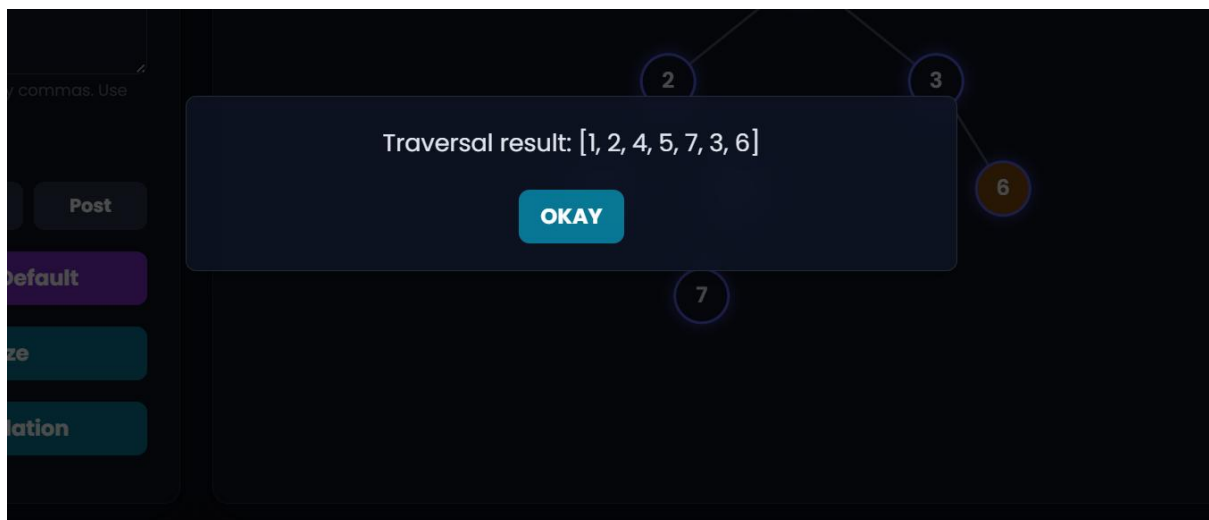
© 2025 Saifullah Bin Yusuf. All rights reserved.

🗨️

🌐

</>

✉️



INORDER

[← Back to Home](#)

 **Tree Traversals**

Input
Input is level-by-level order.

1, 2, 3, 4, 5, null, 6, null, null, 7, null, null, null, null, null

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

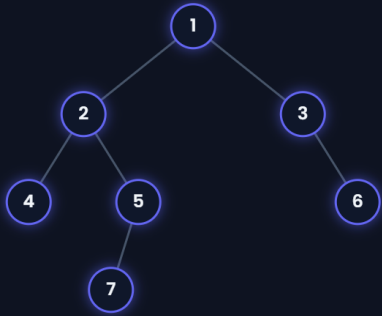
Pre In Post

 **Generate Default**

 **Visualize**

 **Start Simulation**

☐☐☐☐☐☐☐



```
graph TD; 1((1)) --- 2((2)); 1 --- 3((3)); 2 --- 4((4)); 2 --- 5((5)); 3 --- 6((6)); 5 --- 7((7));
```

[← Back to Home](#)

Tree Traversals

Input

Input is level-by-level order.

1, 2, 3, 4, 5, null, 6, null, null, 7,
null, null, null, null, null

Enter numbers separated by commas. Use
null for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

4 2 7 5 1 3 6

© 2025 Saifullah Bin Yusuf. All rights reserved.

Traversal result: [4, 2, 7, 5, 1, 3, 6]

OKAY

POSTORDER

← Back to Home

Input

Input is level-by-level order.

1, 2, 3, 4, 5, null, 6, null, null, 7,
null, null, null, null

Enter numbers separated by commas. Use
'null' for empty nodes.

Traversal Order

Pre

In

Post

 Generate Default

▶ Visualize

▶ Start Simulation

Tree Traversals

1

2

3

4

5

6

7

← Back to Home

Tree Traversals

?

Input

Input is level-by-level order.

1, 2, 3, 4, 5, null, 6, null, null, 7,
null, null, null, null, null

Enter numbers separated by commas. Use
'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

▶ Visualize

▶ Start Simulation

4 7 5 2 6 3 1

```
graph TD; 1((1)) --- 2((2)); 1 --- 3((3)); 2 --- 4((4)); 2 --- 5((5)); 5 --- 7((7)); 3 --- 6((6));
```

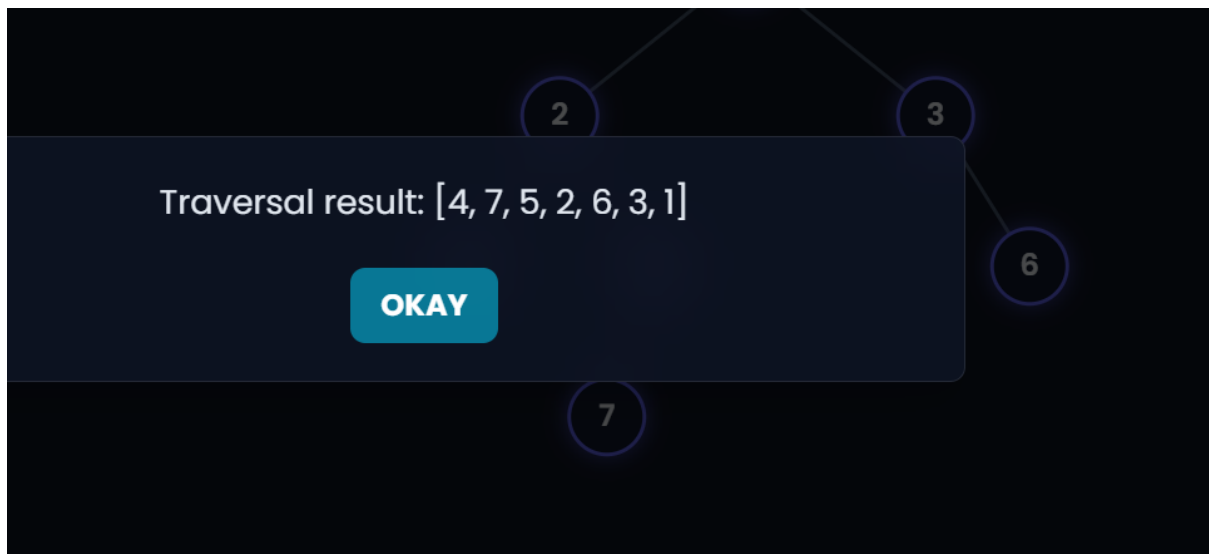
© 2025 Saifullah Bin Yusuf. All rights reserved.

↺

in

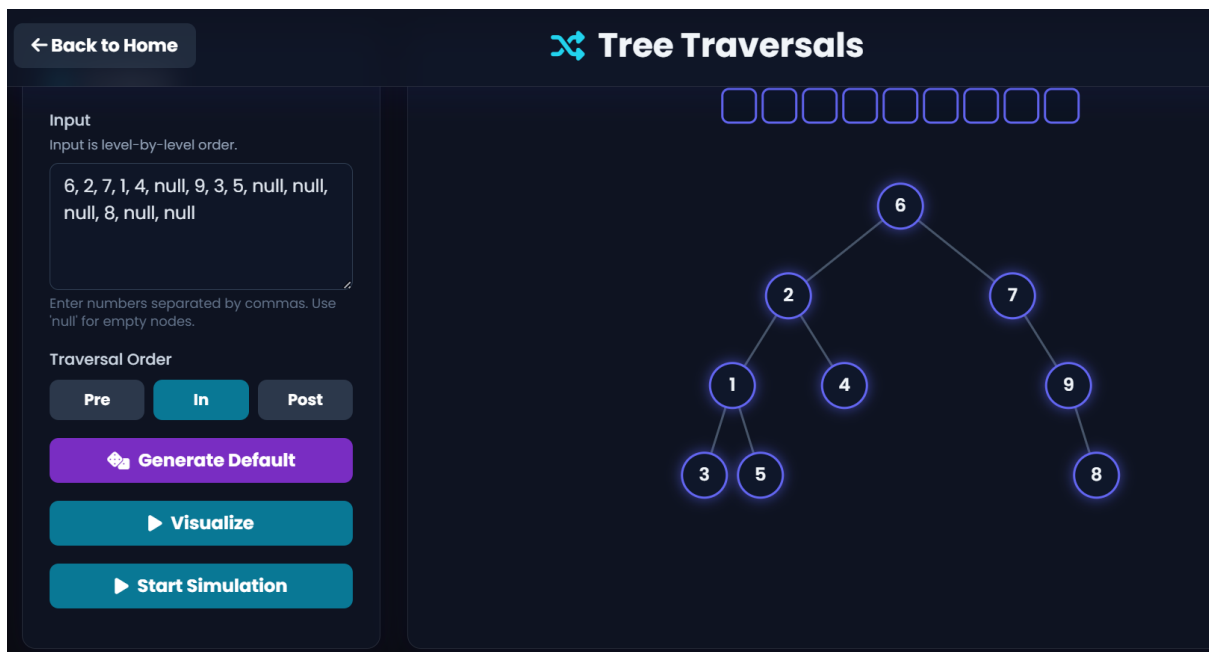
</>

✉



QUESTION 4

INORDER



[← Back to Home](#)

Tree Traversals

Input

Input is level-by-level order.

6, 2, 7, 1, 4, null, 9, 3, 5, null, null, null, 8, null, null

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

3 1 5 2 4 6 7 9 8

6

2

7

1

4

9

8

3

5

© 2025 Saifullah Bin Yusuf. All rights reserved.



2

7

9

8

3

5

Traversal result: [3, 1, 5, 2, 4, 6, 7, 9, 8]

OKAY

3,1,5,2,4,6,7,9,8

A,B,C,D,E,F,G,H,I

QUESTION 5

PREORDER

[← Back to Home](#)

 **Tree Traversals**

Input
Input is level-by-level order.

1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

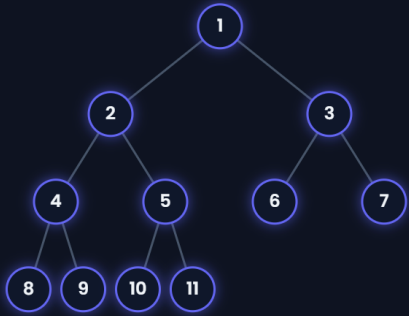
In

Post

 **Generate Default**

 **Visualize**

 **Start Simulation**



© 2025 Saifullah Bin Yusuf. All rights reserved.

[← Back to Home](#)

 **Tree Traversals**

Input
Input is level-by-level order.

1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

 **Generate Default**

 **Visualize**

 **Start Simulation**

1

2

4

8

9

5

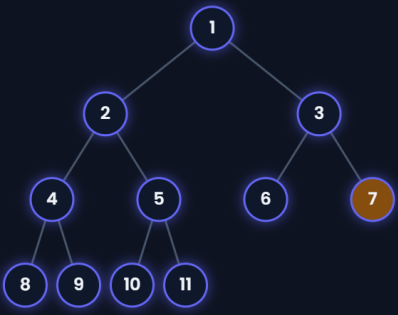
10

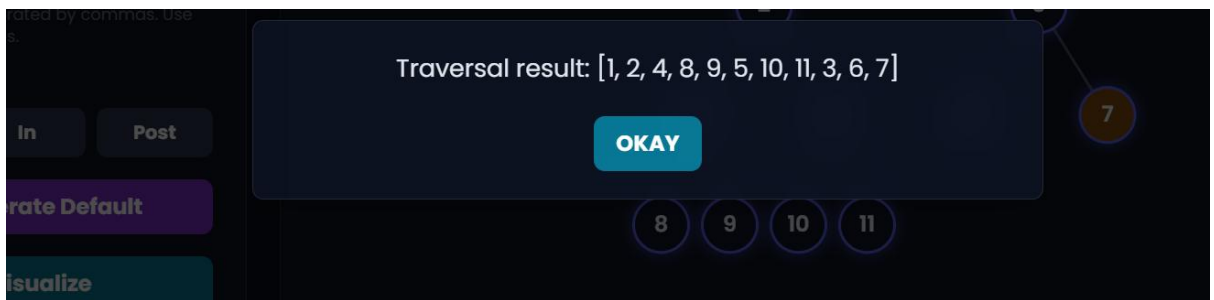
11

3

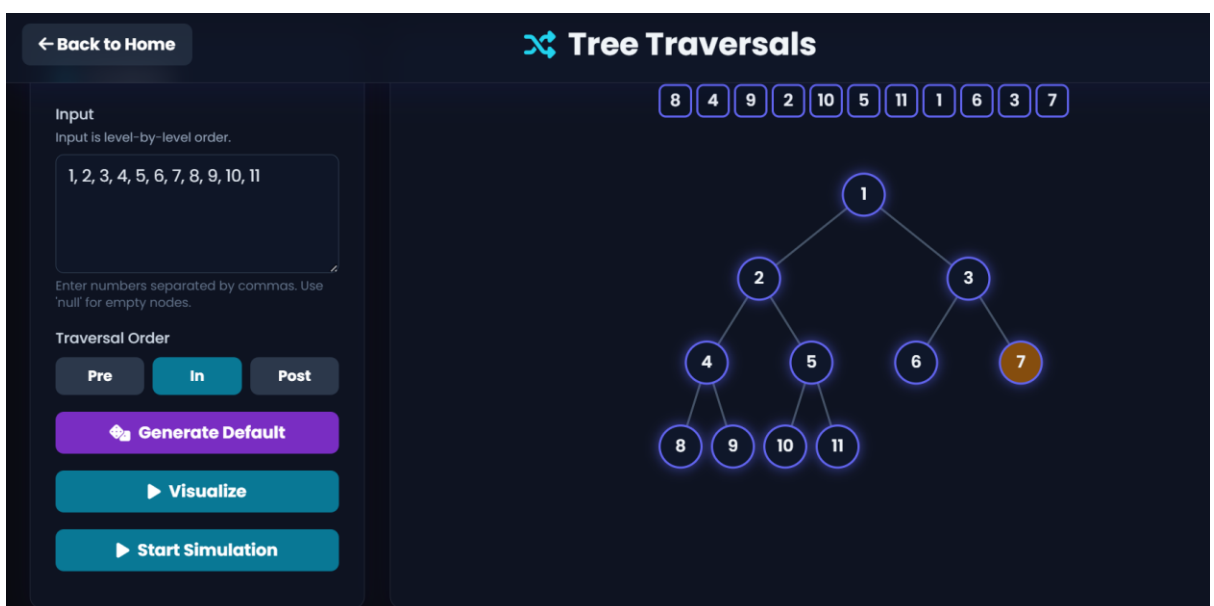
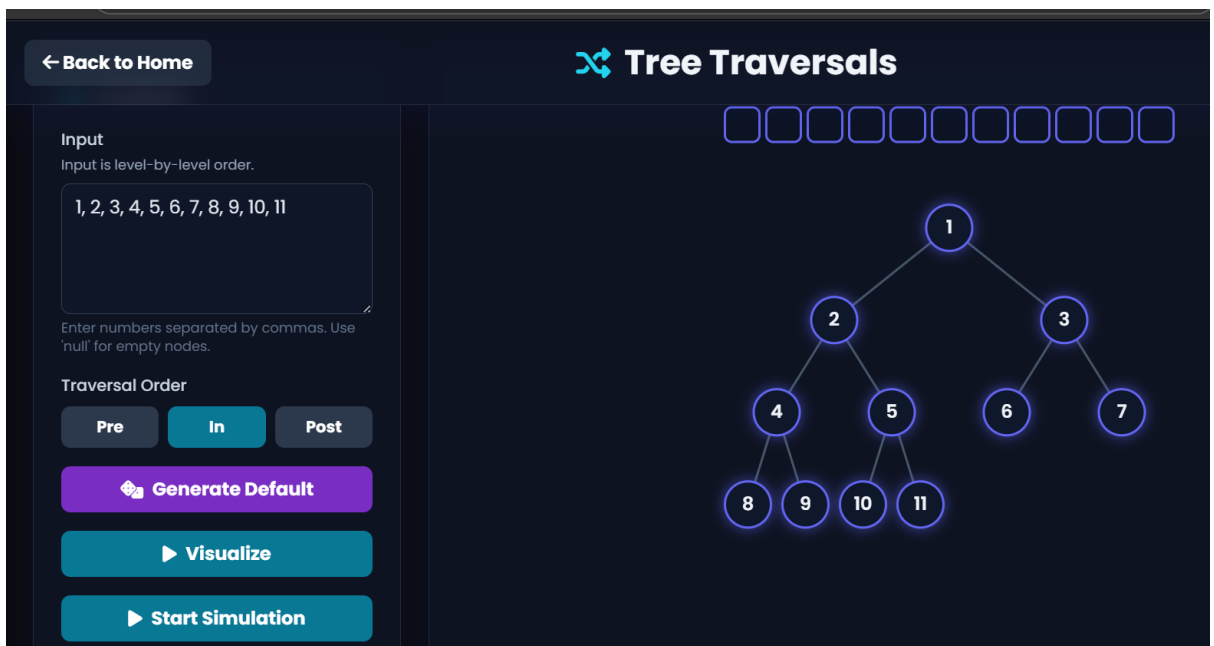
6

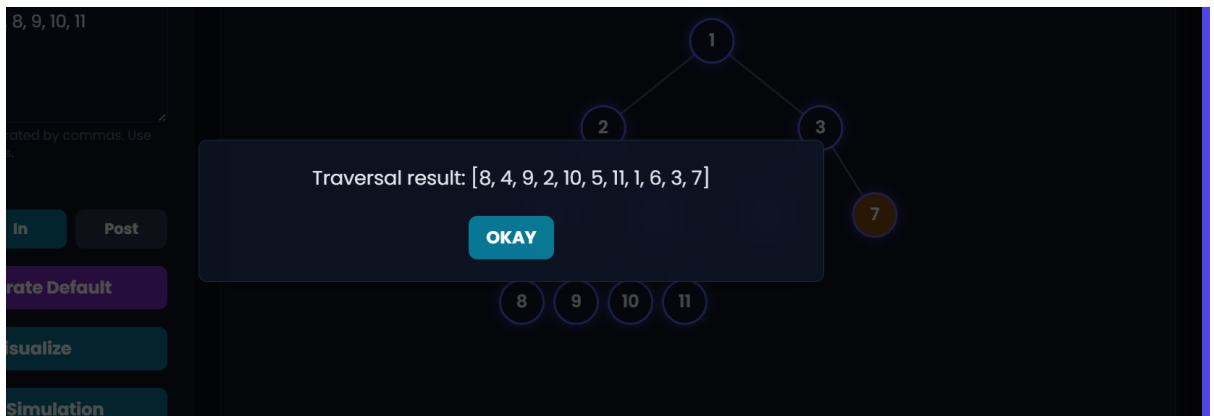
7



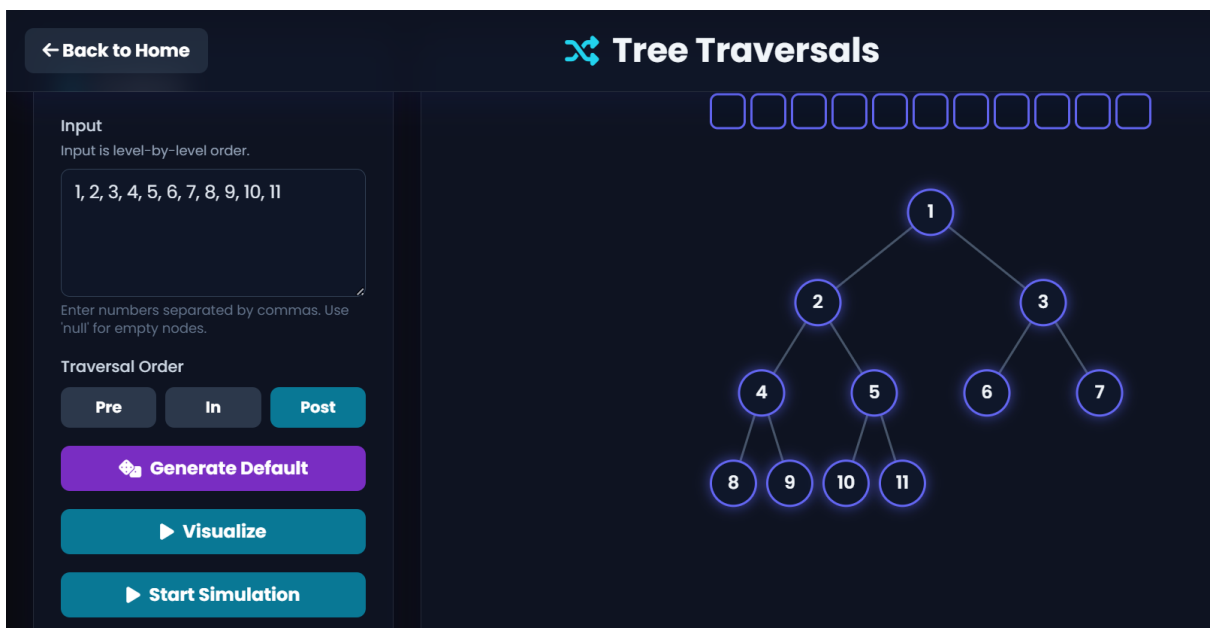


INORDER





POSTORDER



[← Back to Home](#)

Tree Traversals

Input

Input is level-by-level order.

1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

894101152673

1

2

3

4

5

6

7

8

9

10

11

Traversal result: [8, 9, 4, 10, 11, 5, 2, 6, 7, 3, 1]

OKAY